DATA QUALITY ASSESSMENT

Healthcare Operations Dashboard - Encounters Table

1. Overview

For this project I used the encounters table from the Synthea dataset. After cleaning, I had 27,890 records. The main fields I worked with were Id, START, STOP, ENCOUNTERCLASS, PATIENT, BASE_ENCOUNTER_COST, TOTAL_CLAIM_COST, PAYER_COVERAGE, and I created a new column called Duration_Hours.

2. Completeness

I checked for missing values first. There were no blanks in the important columns like START, STOP, ENCOUNTERCLASS and PATIENT. PAYER_COVERAGE was zero in 13,586 rows, which is 48.7% of the data, but I kept it as is because it actually means self-pay or uninsured, not missing data. I also removed duplicate Ids in Power Query, so there are now zero duplicates.

3. Validity and Range

I looked at whether the values made sense. The dates run from 2010 to 2023, so there are no future dates. Costs are all zero or positive, with a maximum of around $15,000 which seems reasonable. PAYER_COVERAGE goes from $0 to about $7,500, which is valid. For ENCOUNTERCLASS I found 6 types — ambulatory, emergency, inpatient, wellness, outpatient and urgent care - all expected. Duration_Hours ranges from 0.1 hours to 180 hours, with 76 cases over 24 hours (0.27%). I verified those are real long-stay inpatient cases.

4. Consistency

I checked if the data logic holds together. TOTAL_CLAIM_COST is greater than or equal to BASE_ENCOUNTER_COST in 98.2% of cases, which makes sense because the claim includes extra fees. In every row, STOP is after START after I filtered out any negative durations. All PATIENT ids also exist in the patients table, so the relationship is intact.

5. What I Did to Clean the Data

In Power Query I did the following steps: I removed duplicates on Id, changed START and STOP to DateTime type, added a custom column Duration_Hours using Duration.TotalHours([STOP]-[START]), and then filtered out any rows where Duration was zero or negative. In DAX I created the measures needed for the dashboard: Total Encounters using COUNTROWS, Total Claim Cost and Payer Coverage using SUM, Zero Coverage by counting where PAYER_COVERAGE = 0, Over 24h Count by counting where Duration_Hours > 24, and % Over 24h by dividing those two.

6. Limitations

There are a couple of things to note. The 48.7% zero coverage makes the average coverage look lower than it really is, so I reported it as a separate KPI instead. The 76 cases over 24 hours are outliers — I kept them for the KPI but I did not include them when calculating average length of stay. Also, this is synthetic data from Synthea, so it is not real hospital data, but it is still useful for showing the process.

7. Final Verdict

Overall the dataset is fit for use. Completeness is 100% on key fields, validity checks all pass, and consistency is over 98%. I validated the final KPIs: Total Encounters 27,890, Total Claim $101.51M, Total Payer $31.10M, Zero Coverage 13,586, and % Over 24h 0.27%. The data is clean and ready for dashboarding.